

## Multiple Choice (10 marks)

1. A particle moves with  $v = \text{const.}$  along the curve  $r = k(1 + \cos \theta)$  (a cardioid). Find  $\ddot{\mathbf{r}} \cdot \mathbf{e}_r = \mathbf{a} \cdot \mathbf{e}_r$ .
  - (a)  $-\frac{v^2}{k}$
  - (b)  $-\frac{3}{4} \frac{v^2}{k}$
  - (c)  $-\frac{7}{4} \frac{v^2}{k}$
  - (d)  $\frac{1}{4} \frac{v^2}{k}$
  - (e)  $-\frac{1}{4} \frac{v^2}{k}$
2. Find the ratio of the focal lengths of a glass lens in water and in air. The refractive indices of the glass and water are 1.5 and 1.33 respectively.
  - (a) 2.50
  - (b) 0.89
  - (c) 2.23
  - (d) 2.75
  - (e) 4.5
3. Find the dimension of the parallelepiped of maximum volume circumscribed by a sphere of radius  $R$ .
  - (a)  $\frac{2}{\sqrt{2}}R$
  - (b)  $\frac{3}{\sqrt{2}}R$
  - (c)  $R$
  - (d)  $\sqrt{2}R$
  - (e)  $\frac{1}{\sqrt{3}}R$
4. A uniformly charged conducting sphere of 1.2 m diameter has surface charge density  $8.1\mu\text{C}/\text{m}^2$ . Find the net charge on the sphere.
  - (a)  $97\mu\text{C}$
  - (b)  $37\mu\text{C}$
  - (c)  $77\mu\text{C}$
  - (d)  $87\mu\text{C}$
  - (e)  $17\mu\text{C}$

5. An airplane with its nose pointing north with an airspeed of 40 km/h in a 30-km/h crosswind (at right angles) has a groundspeed of
- (a) 40 km/h
  - (b) 60 km/h
  - (c) 50 km/h
  - (d) 10 km/h
  - (e) 90 km/h

## True / False (10 marks)

*Answer True or False*

- 6. True or False: If a planet shrinks to one-tenth its original diameter while retaining its mass, your weight would be half of what it was before.
- 7. True or False: In a series circuit, adding more lamps causes the current from the power source to remain constant regardless of the number of lamps.
- 8. True or False: The image of an object placed 100 cm from a plane mirror is located 350 cm from the object.
- 9. True or False: Satellites in orbit are propelled by constant rocket thrust to maintain their trajectory in space.
- 10. True or False: During an isothermal expansion of an ideal gas, 150 J of heat is added to the gas to compensate for the work done on the surroundings.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

- 11. Discuss the factors that determine the intensity of the primary maximum and the first secondary maximum in a diffraction grating with 100 lines, and calculate the ratio of these intensities.
- 12. Calculate the work done in joules when lifting a 5-kilogram mass to a height of 2 meters. Discuss the underlying principles of work and energy involved in this scenario.

*End of Paper*